

JOSHIKARAN.K

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SUMMARY

Backend + AI Engineer specializing in scalable system design and data-driven applications. Experienced in building multi-tenant architectures and high-performance systems. Proficient in Python, JavaScript, and TypeScript, with hands-on experience in frameworks such as FastAPI, React, and NestJS. Adept at rapidly learning and applying new technologies, leveraging AI-assisted tools to accelerate development while maintaining strong ownership of architecture, code quality, and data privacy compliance.

EXPERIENCE

Part Matters

May 2025 – Present

Full-stack Developer

Chennai

- Designed a **modular microservices architecture** (NestJS, FastAPI) separating API, core logic, and AI workloads, enabling scalable and independently optimized system components.
- Built and deployed a **high-performance search and logging stack** using ELK (Elasticsearch), Docker, Redis caching, significantly improving system response times and observability.
- Implemented **multi-tenant RBAC architecture** and modeled complex hierarchical data using adjacency lists and BFS traversal for efficient commodity-to-supplier mapping.
- Developed an **AI-powered assistant** using LLM APIs with validation layers and FAISS-based retrieval for reliable, context-aware conversational workflows.

Locobuzz Solutions, Private Limited

Mar 2024 – June 2024

Python Developer Intern

Mumbai

- Engineered TikTok data retrieval service that processed 5,000+ daily API calls, reducing manual data collection time by 70%
- Automated Amazon/Flipkart review scraping system using Scrapy that collected 50K+ product reviews with 95% accuracy
- Implemented share-link expiration feature that improved platform security by preventing 100% of expired link accesses
- Fixed timezone discrepancies affecting 1,200+ active users, enhancing dashboard reliability

TECHNICAL SKILLS

Programming Languages and Frameworks:	Python, JavaScript, TypeScript, Django, FastAPI, NestJS, Next.js
AI/ML (Applied):	XGBoost, Scikit-learn, PyTorch (inference), Vector Embeddings, FAISS
Databases:	PostgreSQL, MongoDB, Redis, Elasticsearch, SQLite
DevOps:	Git, Docker, CI/CD Pipelines, Azure VM, pm2
Tools:	GitHub Copilot, VS Code

EDUCATION

SRM University of Science and Technology

Sep 2020 – June 2024

B.Tech in Computer Science (AI/ML Specialization)

PROJECTS

AI-Powered Crypto Trading Simulator App - MVP

Jan 2025 – Mar 2025

[GitHub](#) — [Youtube Video Demo](#) — [Documentation](#)

- Built an **AI-powered crypto trading simulator** with a microservices architecture, achieving **78% accuracy** using XGBoost on 2M+ data points and integrating a fine-tuned Mistral-7B + RAG pipeline for real-time sentiment-driven trade recommendations (**↑80% relevance**), complemented by an interactive, user-centric UI/UX.

PCB Defect Detection using YOLOv11

Oct 2023 – Mar 2024

[GitHub](#)

- Developed a **YOLOv11-based PCB defect detection system** achieving **95% accuracy**, trained on **30K+ images** (6 defect classes), with real-time inference (live camera + uploads) reducing inspection time by **60%**, deployed using a Flask backend.

RAGSTRUCT

Feb 2025 – April 2025

[GitHub](#)

- Built a **custom RAG pipeline with key-level (field-wise) retrieval**, converting unstructured text into structured JSON schemas and leveraging FAISS embeddings to reduce token usage by **60%** while improving factual accuracy and minimizing hallucinations through controlled response generation.